

Week 2 Report: Literature Review and Dataset Collection

Project: Analysing the Role of Big Data Analytics for Cyber Threat Detection and Prevention in Banking Organisations of Ireland

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Module: Open Data Practice, Research Practicum

1. What this week covers

In my Gantt chart the literature review runs from August to the end of September, and dataset collection happens in August. So this week I did two things. First, I went back over the 15 papers from Week 1 and compared them with each other, instead of looking at them one at a time. Second, I chose the dataset, downloaded it and checked it properly before any preprocessing starts in September.

2. Literature review progress

When I compared the papers, the first thing I noticed is that tree ensembles keep coming out on top. Random Forest was the best model in Sanda et al. (2026), with 97.2% accuracy, and Dhammayogi and Pramatha (2026) found Random Forest and XGBoost almost level (ROC-AUC 0.992 and 0.995), with SVM further behind at 0.935. Kumar et al. (2025) also report Random Forest ahead of SVM. This supports my plan to compare Decision Tree, Random Forest and SVM, with Random Forest as the model to beat.

The second thing is that the results are hard to compare. The papers use different datasets, different splits and different ways of handling imbalance, and two of them, Naeem et al. (2026) and Ullah et al. (2026), use simulated data. None of the papers compare their models with the simple rules that banks already use, so it is not clear how much machine learning really adds. Because of this I will use one fixed train and test split for every model, report precision, recall, F1 and ROC-AUC, and add a simple rule-based baseline.

Third, big data tools are described more than they are measured. Sathupadi et al. (2025) show Spark and Kafka working with a BiLSTM at up to 1000 Mbps, but most other papers only mention these tools. I cannot build a full Spark cluster in this project, but I can measure how training and prediction time grow as the data gets bigger.

Fourth, robustness is a known concern but rarely tested. Padmanaban et al. (2026) show that making a model harder to fool brings accuracy down to 88.2%, and Kovacevic et al. (2025) warn about data poisoning and evasion attacks. A simple check I can do is to train on earlier months and test on later ones, and to test on merchants the model has not seen.

Finally, explainability and regulation matter for banks. Dhammayogi and Pramatha (2026) and Baisholan et al. (2025) use SHAP to explain predictions, and Laddi et al. (2025) keep data private with federated learning. For an Irish bank, being able to explain why a transaction was flagged is a GDPR requirement, so I will include feature importance in my results.

Table 1 summarises what each paper used and found. Papers marked with * are my baseline papers.

No.	Paper	Method	Data	Main result
1*	Sathupadi et al. (2025)	BiLSTM with Spark and Kafka	Internet banking transactions	98.5% fraud detection accuracy
2	Kumar et al. (2025)	DL, RF, SVM, K-Means	500,000 security incidents	DL 96.8%, RF 94.1%, SVM 92.3%
3	Kokogho et al. (2025)	Review	Fintech	No experiments
4*	Pospichal et al. (2026)	RF with feature selection, SMOTE	UNSW-NB15	F1 0.93 to 0.98
5	Baisholan et al. (2025)	RF and XGBoost ensemble, SHAP	European card data	Recall 95%, AUC-PR 97%
6	Padmanaban et al. (2026)	RL with adversarial training	Simulated attacks	88.2% accuracy, 5.1% false positives
7*	Hozouri et al. (2025)	Survey of IDS	CIC-IDS2017, KDDCup99, UNSW-NB15	No experiments
8	Laddi et al. (2025)	Federated anomaly detection	Kaggle network intrusion	Privacy-preserving IDS
9	Zhang (2025)	Ensemble of 21 DL models	BoT-IoT	99.985% accuracy
10	Naeem et al. (2026)	LR, SVM, RF, LightGBM, XGBoost	Simulated, 400 rows	XGBoost 98.2% (template only)
11	Dhammayogi and Pramatha (2026)	RF, XGBoost, SVM with SHAP	Digital banking fraud	ROC-AUC 0.995, 0.992, 0.935
12*	Sanda et al. (2026)	LR, RF, SVM, GB, NN with SMOTE	Kaggle banking fraud	RF 97.2%, AUC 98.4%
13	Eshra et al. (2025)	Threat intelligence study	Financial institutions	33.7% lower breach likelihood
14	Ullah et al. (2026)	XGBoost and LSTM hybrid	Simulated, 125,000 rows	96.4% accuracy, ROC-AUC 0.982
15	Kovacevic et al. (2025)	Discussion paper	Banking	Adversarial AI risks

Table 1. Summary of the reviewed papers.

From this comparison I have narrowed my research gaps to five points. There is no comparison with simple rules. Results use different setups and cannot be compared. Big data claims are rarely measured. Changes over time and attacks are rarely tested. And there is very little Irish or EU regulatory context. Each of these is covered by a specific test in my plan.

I also noticed that most of these studies use network traffic datasets such as UNSW-NB15 and BoT-IoT, or simulated data. Network traffic is not the same as bank transactions, and simulated data cannot show real performance. This is why I wanted an open dataset of labelled bank transactions.

3. Dataset collection

Before choosing, I wrote down what I needed: bank transaction records, a label for suspicious or fraudulent activity, an open licence, a size I can train on my laptop, and no personal data. I looked at four options. The Bank Account Fraud dataset (Jesus et al., 2022) is about account applications, not transactions. PaySim (Lopez-Rojas et al., 2016) is mobile money. CICIDS2017 is network traffic. The Kaggle dataset "Bank Transaction Records along Suspicious Flags", which I named in my proposal,

meets all five points, so I chose it.

I downloaded it with the Kaggle API. Kaggle lists the licence as MIT and describes the data as a simulation of bank transactions. The file is transactions_Dataset.csv (2.13 MB). I saved its SHA-256 checksum so I can show later that I used the same file throughout: 3863ecc55dd883f959c79362de9a2425ad8ac0ec05e5d6d71890488fc65d83f1

The dataset has 33,611 rows and 6 columns. Table 2 describes each column.

Column	Type	Unique values	Range	Meaning
date	object	757	text	Transaction date (dd-mm-yyyy string, parsed to datetime in preprocessing)
description	object	43	text	Channel / merchant code, e.g. NETTXN:EBAY (categorical text)
debit	float64	18,521	0.0 to 1793799.36	Amount leaving the account (0 when the row is a credit)
credit	float64	15,083	0.0 to 1993110.41	Amount entering the account (0 when the row is a debit)
balance	float64	33,611	-720071.17 to 22218339.63	Account balance after the transaction
isSuspicious	int64	2	0.0 to 1.0	Target label: 1 = flagged suspicious, 0 = normal

Table 2. Data dictionary.

I then checked the data quality. There are no missing values and no duplicate rows. All dates follow the day-month-year format and run from 2022-12-16 to 2025-01-12. Debit and credit are never negative, and every row has either a debit or a credit, never both. 310 rows have a negative balance, which I read as overdrafts. The balance is different on every row and only matches the previous row's balance in 2.15% of cases, so the rows are not one running account. Because of that I will test the models with and without the balance column.

14,203 of the 33,611 transactions (42.3%) are marked as suspicious. That is much more balanced than real bank fraud, where the rate is well under 1%. It means class weights should be enough to handle the imbalance, but it also means I should not treat the number of alerts as realistic.

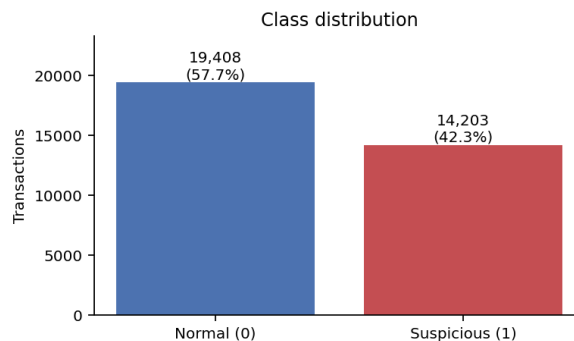


Figure 1. Normal and suspicious transactions.

The most important thing I found is how much the description column matters. 26 of the 43 descriptions are never flagged. These are payment channels such as PCA, IMPS and cash deposits.

Online shops, ticket booking sites and crypto exchanges are flagged in about 70 to 75% of their rows (Figure 2). So a model could learn the labelling rule instead of real threat behaviour. To check this I will add a test where some descriptions are kept out of training.

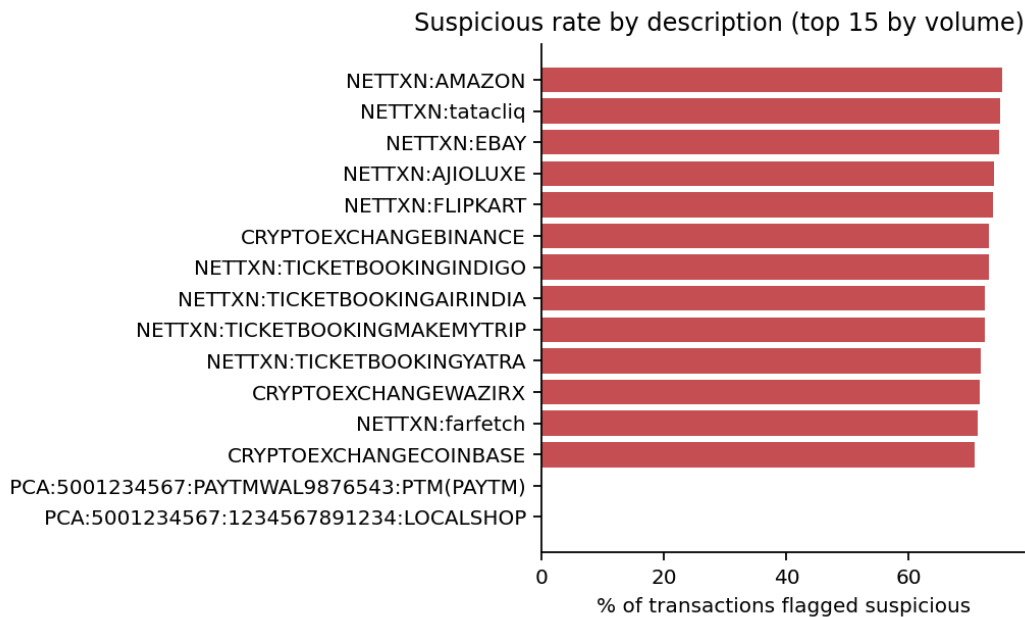


Figure 2. Share of suspicious transactions for the 15 most common descriptions.

Suspicious transactions are also bigger. For debits, the median suspicious amount is 571,350 compared with 115,298 for normal ones. Amounts vary a lot, so I will use a log transform (Figure 4). The data is also not spread evenly over time. A normal month has about 724 rows, but 2023-03 has 10,637 rows and a much higher suspicious rate (Figure 3). This is another reason to test on later months as well as on a random split.

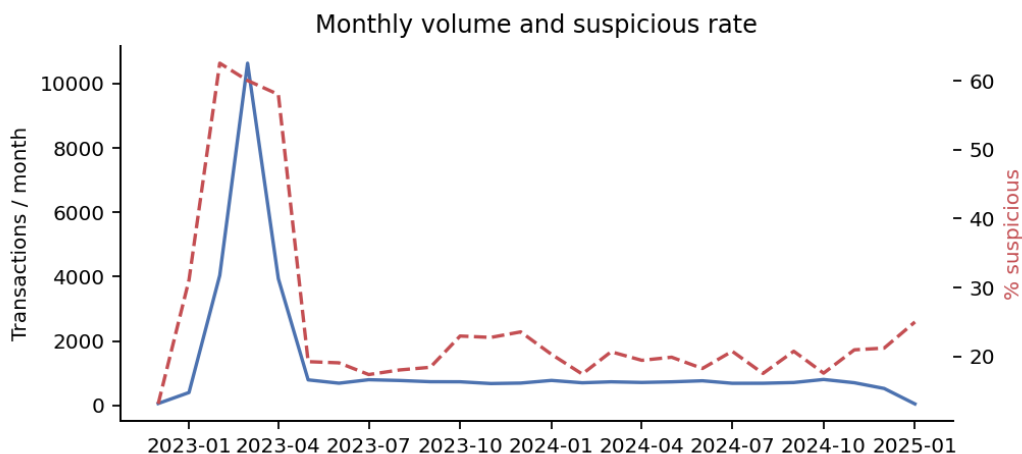


Figure 3. Transactions per month (solid line) and suspicious rate (dashed line).

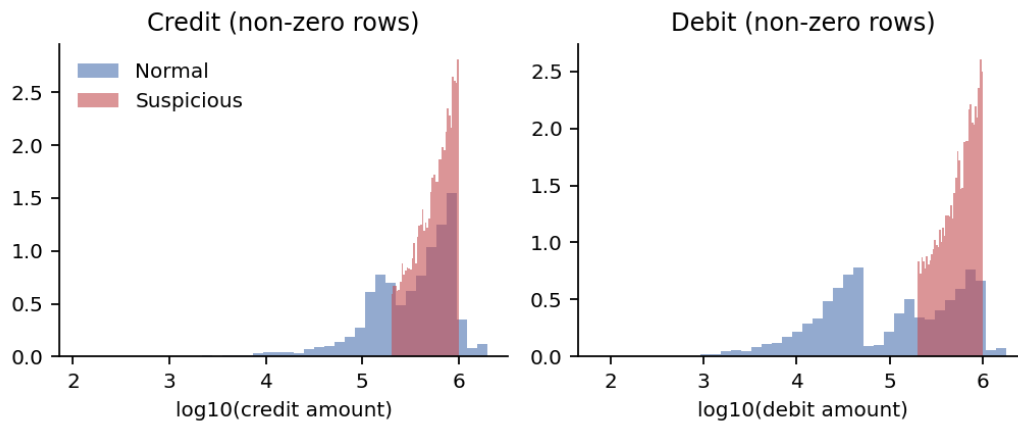


Figure 4. Credit and debit amounts on a log scale.

4. Ethics

The data is simulated and has no names, real account numbers or other personal details, so GDPR does not apply to the raw file. The account-style numbers inside some descriptions are placeholders. The MIT licence allows academic use. I keep the file on my own machine and do not upload it anywhere. Since the data is simulated and not Irish, I will present my results as a test of the methods, not as a measurement of Irish banks.

5. Next week

In Week 3 I will keep reading, mainly papers on fraud patterns changing over time and on rules compared with machine learning. I will also start preprocessing: converting the dates, splitting the description into channel and merchant, log-transforming the amounts, and creating one fixed 80/20 train and test split that I will use for all experiments.

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